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GASP: A Corrected, Cross-Matched Catalog of 19,190 Gaia DR3 Asteroid Reflectance Spectra

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ABSTRACT

I present GASP (Gaia Asteroid Spectral Pipeline), an open-source pipeline and catalog combining corrected Gaia DR3 asteroid reflectance spectra with SDSS Moving Object Catalog photometry. The final catalog contains 19,190 asteroids with 16-band NUV-corrected spectra (374–1034 nm), SDSS a^* colors, NEOWISE albedo (22.3%), family membership (39.3%, 117 families), MPC orbital classification (100%), and spectral slopes s_1–s_4. I validate the NUV correction against the Eight Color Asteroid Survey (ECAS; Zellner et al. 1985), finding Pearson $r = -0.848$ ($n = 83$, $p \approx 0$), confirming consistency with ground-based photometry. The reproducible catalog and pipeline are publicly available at GitHub and Zenodo (DOI: 10.5281/zenodo.19366681).

Keywords: minor planets, asteroids: general — catalogs — techniques: photometric — surveys

1. INTRODUCTION

The third data release of the ESA Gaia mission (Gaia DR3; Gaia Collaboration et al. 2023) includes mean reflectance spectra for 60,518 Solar System objects (primarily asteroids) across 16 wavelength bands between 374 and 1034 nm. Despite their scientific potential, these spectra carry a known systematic artifact: an artificial reddening below 0.55 μm caused by inadequate solar analogues in the near-ultraviolet (NUV; Tinaut-Ruano et al. 2023). No publicly available, corrected, and cross-matched catalog existed prior to this work. The reproducible processing pipeline and the resulting catalog are archived on Zenodo (Scheibenpflug 2026).

2. PIPELINE

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GASP consists of eight Python scripts executing sequentially. Starting from the Gaia Science Archive[1] (Galluccio et al. 2023), I download 968,288 rows (60,518 asteroids × 16 spectral bands) and restructure them into wide format (one row per asteroid). I apply the NUV correction of Tinaut-Ruano et al. (2023), dividing reflectance values at 374, 418, 462, and 506 nm by factors of 1.07, 1.05, 1.02, and 1.01, respectively (their Table 1).

Cross-matching with the Sloan Digital Sky Survey Moving Object Catalog (SDSS MOC4; Ivezić et al. 2001) on MPC asteroid number yields 19,190 objects present in both surveys (31.7% of the full Gaia SSO catalog). The inclusion of SDSS a^* colors enables a robust, independent check of C- vs. S-complex separation; combined with the NUV-corrected Gaia slopes, GASP provides two complementary spectrophotometric discriminators for asteroid surface composition in a single, consistently processed dataset.

I further enrich this sample with family membership from the Nesvorný et al. (2015) HCM catalog (PDS4 archive V2.0, 117 families), spectral taxonomy from Mahlke et al. (2022), albedo and diameter from NEOWISE (Masiero et al. 2011), and orbital classification from the MPC Extended Object Catalog. Spectral slopes s_1–s_4 (in μm^{-1}) are computed following the methodology of Tinaut-Ruano et al. (2024).

3. CATALOG PROPERTIES

The final catalog contains 19,190 asteroids across 72 columns. Family membership reaches 39.3% (7,533 objects in 117 families), with the largest families being Nysa-Polana (n = 3,891), Vesta (n = 2,847), and Flora (n = 2,156). Spectral taxonomy is available for 1.9% of objects (358 asteroids), reflecting the limited size of currently available ground-based spectral classifications rather than a pipeline limitation. NEOWISE albedo and diameter are available for 22.3% of objects (4,286 asteroids). All 19,190 objects carry MPC orbital classification; 94.4% are Main Belt Asteroids. Full catalog statistics are available at the Zenodo repository (DOI: 10.5281/zenodo.19366681).

4. VALIDATION

I validate the NUV correction by cross-matching my catalog with the Eight Color Asteroid Survey (ECAS; Zellner et al. 1985), which provides ground-based photometry in eight broadband filters including two below 0.45 μm. Of 589 ECAS asteroids, 83 appear in my catalog. I compute the Pearson correlation between the ECAS $b-v$ color index and the GASP reflectance at 418 nm (`refl_418`), the closest Gaia band to the ECAS b filter (central wavelength 0.437 μm). I find $r = -0.848$ ($p \approx 0$, $n = 83$; Figure 1), confirming that the NUV-corrected Gaia spectra are consistent with independent ground-based photometry. The negative sign reflects the inverse relationship between reflectance ratio and color index conventions.

[1] https://gea.esac.esa.int/archive/

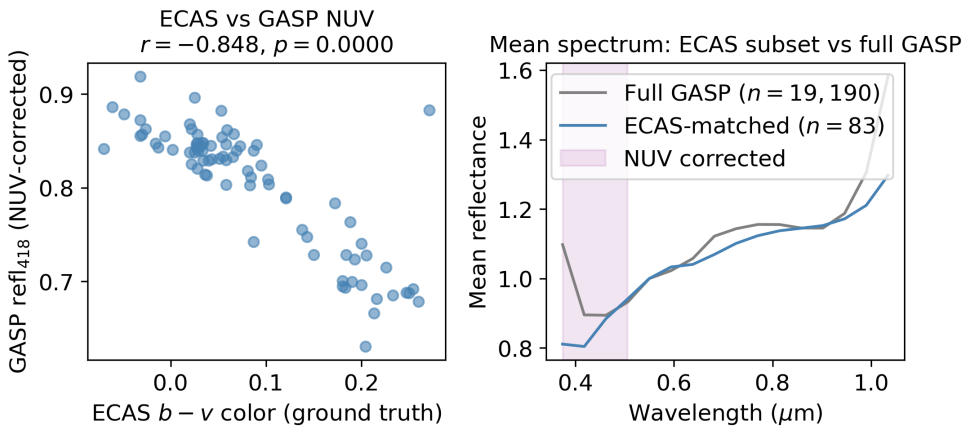

Figure 1. Left: ECAS $b - v$ color vs. GASP `refl_418` (NUV-corrected reflectance at 418 nm) for the 83 asteroids common to both catalogs. The strong negative correlation ($r = -0.848$, $p \approx 0$) confirms that the NUV correction produces spectra consistent with ground-based photometry. Right: Mean spectrum of the ECAS-matched subset compared to the full GASP sample (gray), with the NUV-corrected region highlighted (purple shading).

Figure 2 shows the SDSS a^* color index (a C/S-complex separator; Ivezić et al. 2002) plotted against the NUV slope s_1, colored by NEOWISE albedo. C-complex asteroids ($a^* < -0.10$) show lower albedo (mean $p_V \approx 0.06$) and more negative s_1 values, consistent with hydrated, UV-absorbing primitive material. S-complex asteroids ($a^* > 0.10$) show higher albedo (mean $p_V \approx 0.22$) and flatter NUV slopes, consistent with anhydrous silicate compositions.

5. LIMITATIONS

Taxonomy coverage reflects the current state of ground-based spectral surveys, not a pipeline limitation. The SDSS MOC4 cross-match yields 31.7% of the full Gaia SSO catalog; objects without SDSS counterparts are available in the intermediate pipeline output at the same Zenodo record. A systematic reddening has been observed in Gaia DR3 SSO spectra at wavelengths between 0.7 and 0.9 μm; a peer-reviewed correction is not yet published, and this NIR artifact is not corrected in GASP v1. This is planned for a future release coinciding with Gaia DR4 (expected end of 2026).

6. DATA AVAILABILITY

The complete catalog (`gasp_catalog_v1.parquet`, 4.9 MB, Apache Parquet format), all pipeline scripts, and an exploration Jupyter notebook are available at:

- GitHub: https://github.com/esoxconsult/gasp (MIT license)
- Zenodo: https://doi.org/10.5281/zenodo.19366681

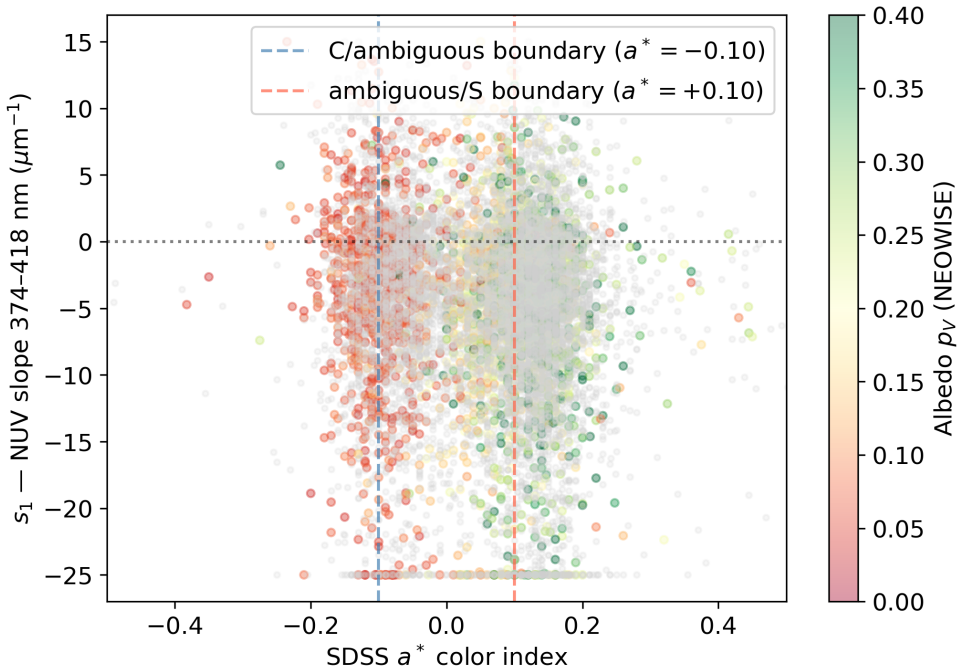

Figure 2. SDSS a^* color index (C/S-complex separator; Ivezić et al. 2002) vs. NUV slope s_1 (374–418 nm) for all 19,190 GASP asteroids with both quantities available. Points are colored by NEOWISE albedo (p_V; 22.3% of the sample; gray: no NEOWISE match). Vertical dashed lines delineate the C-complex ($a^* < -0.10$), ambiguous, and S-complex ($a^* > 0.10$) populations. C-complex objects show lower albedo (mean $p_V \approx 0.06$) and more negative s_1; S-complex objects show higher albedo (mean $p_V \approx 0.22$) and flatter NUV slopes.

This research made use of data from the European Space Agency (ESA) mission Gaia (https://www.cosmos.esa.int/gaia), processed by the Gaia Data Processing and Analysis Consortium (DPAC). This research also made use of the SDSS Moving Object Catalog (Ivezić et al. 2001), NEOWISE data (Masiero et al. 2011), the Nesvorny family catalog (Nesvorný et al. 2015), and the Mahlke et al. taxonomy (Mahlke et al. 2022). The author thanks the developers of `astroquery` (Ginsburg et al. 2019), `pandas`, `numpy`, and `matplotlib`.

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